ReliefLink: Complete Master Capstone Presentation Script

Exhaustive Step-by-Step Live Demonstration Covering All Web and Mobile Subsystems

Target Presentation Duration: 12 – 15 Minutes (Comprehensive Defense)

Target Audience: Capstone Examination Panel, Advisers, Dean

Demonstration Platforms: Flutter Mobile (iOS/Android) + React Admin Portal (Web)

Underlying Blockchain: Hyperledger Besu Private Ethereum Network (QBFT)

Payment Gateway: PayMongo Automated API (QR Ph, GCash, E-Wallets)

Decision Support: Multi-Criteria Need-Weighted Prescriptive Analytics Engine

ACT I: EXECUTIVE INTRODUCTION & THE CORE PROBLEM

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Visual Display: Title Slide on presentation screen or ReliefLink Web Landing Page.
Presenter Action: Stand confidently, establish eye contact with the capstone committee, and deliver the opening hook.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Good morning, respected chair and members of the capstone panel. Today, we are proud to present ReliefLink: A Blockchain-Powered Parish Relief and Donation Management System with Prescriptive Decision Support.

Charitable giving in religious and community organizations is built on goodwill. However, our research revealed a critical vulnerability: once donations enter church collection boxes or bank accounts, donors have zero post-donation visibility. They never know when, where, or how their contributions were actually used. Furthermore, parish administrators lack scientific optimization tools, allocating resources by guesswork.

ReliefLink solves this with an end-to-end ecosystem: a seamless mobile application for donors, a comprehensive administrative portal with prescriptive analytics for church councils, and an enterprise Hyperledger Besu private blockchain guaranteeing mathematical, tamper-proof transparency."

PART 1: MOBILE APPLICATION MODULES (DONOR-FACING)

Module M1 & M2: Splash Screen, Onboarding, and Authentication

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Mobile App Splash Screen with animated ReliefLink logo transitioning to Login Screen.
Action: Show the login form. Point out the password visibility toggle and form validation. Mention the Register Screen and Forgot Password OTP verification flow. Log in with donor account: ikathpie3@gmail.com.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"We start with the donor's mobile experience. The app begins with an animated splash screen and securely initializes state. Parishioners authenticate via JWT tokens. The mobile client includes full account creation, input validation, and email OTP password recovery. When we log in as our active donor, their session is persisted locally using encrypted shared preferences."

Module M3: Home Dashboard, Giving Stats & Fluid Momentum Physics

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Home Screen (Tab 0).
Action: Scroll up and down with the mouse/touchpad to showcase the smooth bouncing momentum scroll physics (`BouncingScrollPhysics`) and drag-to-scroll support. Point out the time-aware greeting ('Good morning/afternoon'), the 'Support Sto. Domingo Church' Hero banner, and the 'Your Giving' KPI cards showing **TOTAL DONATED: ■160.00** and **TRANSACTIONS: 7**.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Arriving at the Home Dashboard, notice the fluid user experience: our application implements native bouncing momentum physics and full mouse drag support. At the top, donors see a warm, time-aware greeting and our Support Sto. Domingo Church banner.

Crucially, the 'Your Giving' section immediately reassures the donor by showcasing their total verified lifetime contributions—here ■160.00 across 7 transactions. From this dashboard, donors can quickly jump to make a donation or explore full transparency reports."

Module M4: Cashless Donation Gateway & PayMongo Integration (Donate Tab)

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Bottom Navbar → Tap the prominent center **Donate** button.
Action: Show the destination selector: choose **'Parish General Fund'** (or *Elderly, Student Scholarships, Disaster Aid*). Select a preset amount of **■25.00** or enter a custom amount. Tap **'Proceed to PayMongo Checkout'**. Display the simulated QR Ph / GCash window.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Next is Module M4: Cashless Donation. By tapping the prominent center Donate button, parishioners can give without physical cash handling risks. Donors have the freedom to earmark their donation for a specific sector—such as Scholarships, Elderly Healthcare, or Calamity Relief.

*Through our native **PayMongo** integration, the app generates automated QR Ph and GCash checkout sessions. When payment completes, PayMongo's secure webhooks notify our backend immediately, verifying the donation without requiring manual receipt uploads."*

Module M5: Donation Summary, 5-Second Silent Sync & Blockchain Tx Proof

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Bottom Navbar → Tap **History** (Donation Summary).
Action: Point out that the list automatically refreshes every 5 seconds in the background without screen flickering. Tap the top verified donation card (■25.00) to expand its details. Scroll down inside the card and highlight the **Blockchain Information** box with the 0x... hash. Tap the **Copy** button to show the 'Blockchain hash copied to clipboard' snackbar.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

*"Here in Module M5: Donation Summary, our app runs a **5-second silent background polling timer**. Whenever a donation is verified or an admin disburses funds, the UI updates automatically without interrupting the user.*

*Notice what happens when we expand this receipt: at the bottom is the **Blockchain Information** box displaying a unique 0x hexadecimal hash. This is the donor's cryptographic proof, permanently etched on our Hyperledger Besu smart contract. The donor can copy this hash anytime to verify that their transaction exists immutably on the distributed ledger."*

Module M6: Reports & Transparency — 'My Contributions' & Impact Breakdown

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Open **Reports & Transparency** from Home or drawer.
Action: Scroll down to demonstrate the NestedScrollView: the blue overview card smoothly scrolls up while the 3 tabs pin neatly below the AppBar. Focus on the **'My Contributions'** tab. Point out the utilization progress bar: **Disbursed: ■45.00 (28%)** vs **Cash on Hand: ■115.00 (72%)**. Tap open the donation card to reveal the **GIVING IMPACT BREAKDOWN**: show the event *'Community Kitchen Feeding Program'*, ■25.00, notes: *'Purchased warm meals for 15 children'*, and remaining balance: ■0.00. Briefly switch to the **'Parish Expenses'** and **'Need Allocation (%)'** tabs.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"This is the heart of our accountability system: Module M6: Reports and Transparency.

Notice our sticky nested header layout: the overview card glides out of the way, pinning the tabs so the user has the entire screen to inspect records.

*Under our newly refined **'My Contributions'** tab, donors see exactly what happened to their money: out of ■160.00 given, ■45.00 has been deployed to relief operations, while ■115.00 remains safely in the treasury.*

*When we expand a donation, the **Giving Impact Breakdown** details the exact outreach mission: ■25.00 disbursed for the 'Community Kitchen Feeding Program' on October 14, with operational notes confirming 15 children received warm meals. We also provide full transparency into all **Parish Expenses** and sector-by-sector **Need Allocation percentages**."*

Module M7 & M8: Parish Bulletins, Profile Management & Settings

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Bottom Navbar → Tap **Announcements** (Tab 1), then tap **Profile** (Tab 4).
Action: In Announcements, show the category dropdown filter (All, General, Urgent Relief). In Profile, show user details (Name, Email, Role), 'Edit Profile', 'Change Password', 'About ReliefLink', and Logout.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Rounding out the mobile app: Module M7 provides categorized parish announcements so parishioners stay informed about upcoming masses and urgent disaster mobilization.

Finally, Module M8 gives parishioners complete profile control, password modification with validation, and an About ReliefLink dialog outlining the parish mission."

PART 2: WEB ADMINISTRATIVE PORTAL (PARISH-FACING)

Module W1 & W2: Superadmin Authentication & Executive Solvency Analytics

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Switch display to Chrome Browser showing React Web Portal (localhost:3000).
Action: Log in as Superadmin (Francis Arillo). Show the top KPI summary: Total Collected, Total Beneficiaries, Expenses Disbursed, and Treasury Reserves. Scroll past the inflow/outflow charts and recent transaction feed.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Now, we transition to the Web Administrative Portal, designed for parish priests, finance councils, and ministry heads. Upon superadmin authentication, the Executive Dashboard delivers real-time visibility over church solvency: monitoring total collections, active beneficiary rolls, and historical disbursement trends."

Module W3 & W4: Donation Verification & Relief Mission Fund Disbursement

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Sidebar → Click **Donation Management**.
Action: Show the full donation ledger with filter tabs (All, Pending, Approved, Cash on Hand, Partially Used, Fully Used, Canceled). Click on an approved donation (e.g., ■30.00 General Fund). In the modal, click **Edit Status / Disburse**. Select status: **'partially used'**, enter Disbursed Amount: **20.00**, Purpose: **'Typhoon Relief Operation'**, Notes: **'Purchased 5 relief food packs for affected families'**. Click **Save**. Show the instant balance update.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"In Module W3: Donation Management, administrators oversee physical and digital inflows. When the parish conducts an outreach mission, we use Module W4: Fund Disbursement. The admin selects a verified donation, chooses 'Partially Used', and allocates ■20.00 to the 'Typhoon Relief Operation' with operational notes.

The system automatically computes the remaining balance of ■10.00. The moment this is saved, our backend updates the donor's mobile screen within seconds, creating the seamless transparency link we witnessed earlier."

Module W5: Beneficiary Demographic Registry & Scholar Grant Management

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Sidebar → Click **Beneficiary Management**.
Action: Show the beneficiary roster filtered by sector (Student Scholars, Indigent Families, Senior Citizens & PWD, Calamity Victims). Click on the **Scholar Grant Renewal Matrix** to show how scholars are tracked by GPA, attendance, and community service hours.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Module W5 manages the parish's verified beneficiaries. Beneficiaries are categorized into sectors—such as Scholars, Indigent Families, and Seniors. For student scholars, our system includes a multi-criteria grant renewal matrix that tracks academic compliance and church service hours, ensuring scholarship stipends are awarded equitably."

Module W6: Prescriptive Decision Support Engine & Optimization Matrix

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Sidebar → Click **Reports**.
Action: Point out the top 4 cards: **OPTIMIZATION BASELINE (171 Members, ■161,343 Pool)**, **PRESCRIBED AID TARGET**, **ACTUAL AID DISBURSED**, and **REMAINING ALLOCATION GAP (■142,420.00)**.
Scroll through the table: highlight **Ministry Sector**, **Urgency Multiplier** (1.6x Scholars, 1.4x Indigent, 1.3x Seniors, 1.2x Calamity, 1.0x Livelihood), **Target Share**, **Prescribed Target**, **Variance / Gap**, and **Prescriptive Directive**. Point out the blue **'Manage Aid'** button.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"Here is our most sophisticated administrative tool: Module W6: The Prescriptive Decision Support Engine.

Traditional church councils guess how to divide funds. ReliefLink replaces guesswork with mathematics. Our engine takes the 171 evaluated members and applies sector urgency multipliers: 1.6x for scholars facing tuition cutoffs, 1.4x for indigent nutrition, and 1.3x for senior healthcare.

It prescribes the exact target budget and calculates the 'Remaining Allocation Gap'. If a sector is underfunded, it outputs clear prescriptive directives like: 'Disburse ■30,222.00 to reach optimal 100% allocation'. By clicking 'Manage Aid', the administrator is guided directly to disburse the funds and close the gap."

Module W7 to W10: Public Expense Auditing, Documents, Bulletins & User Roles

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Screen: Briefly navigate through **Documents**, **Announcement**, and **User Management** tabs.
Action: Show the uploaded receipts/vouchers in Expenses/Documents, the announcement composer, and the staff role permissions matrix.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"The administrative suite is completed by Module W7: Public Expense Auditing where receipts are uploaded for donor inspection, Module W8: Document Repository for financial audit archiving, Module W9: Church Bulletin management, and Module W10: Role-Based User Management enforcing strict access boundaries between Superadmin, Staff, and Viewers."

PART 3: HYPERLEDGER BESU BLOCKCHAIN SUBSYSTEM

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Visual Display: Terminal showing Besu node RPC logs or VS Code displaying DonationRegistry.sol smart contract.
Action: Highlight function recordDonation(address, donorName, amount, refNum, blockHash) and explain zero-gas consensus.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

*"Powering this entire architecture is our private **Hyperledger Besu Ethereum blockchain**.*

Unlike public Ethereum networks that charge unpredictable gas fees, Hyperledger Besu operates on an enterprise Proof-of-Authority consortium model with zero gas price (`min-gas-price = 0`). The diocese and participating parishes run validator nodes without financial overhead.

Every approved transaction executes our DonationRegistry.sol smart contract, minting an immutable cryptographic block. If an unauthorized actor tampers with MongoDB to alter amounts or fake expenses, the off-chain data immediately fails hash reconciliation with the blockchain, mathematically exposing any fraud."

PART 4: CONCLUSION & PANEL DEFENSE Q&A; SHEET

■■ LIVE SCREEN DEMO ACTION & WHAT TO SHOW:

Visual Display: Split-screen showing Flutter Mobile alongside Web Dashboard, or final summary slide.
Action: Stand straight, thank the panel, and transition to open questioning.

■■ EXACT SPOKEN WORDS (SPEAKER SCRIPT):

"In conclusion, ReliefLink elevates parish charity into the modern digital age. By merging automated mobile cashless giving, prescriptive allocation modeling, and private blockchain immutability, we transform charitable giving from a blind leap of faith into an audited, transparent partnership between parishioners and the church.

Thank you very much, and we welcome your questions and feedback."

Master Panel Q&A; Defense Strategy Sheet

Anticipated Panel Question	Recommended High-Scoring Technical Answer
1. Why use a blockchain instead of a secure relational database like PostgreSQL?	Even the most secure traditional database has superusers (DBAs) with root permissions to execute silent UPDATE or DELETE queries. A blockchain is append-only: once a block hash is computed and linked, historical data cannot be edited without breaking the entire chain's consensus, providing mathematically provable tamper evidence.
2. Do ordinary churchgoers need crypto knowledge or MetaMask?	No. Donors pay through standard Philippine payment rails (GCash, Maya, QR Ph via PayMongo). The blockchain works invisibly behind the scenes in the Node.js backend, exposing only the verifiable transaction hash on their digital receipt.
3. What makes your Prescriptive Analytics 'prescriptive' rather than descriptive?	Descriptive analytics merely tells you how much money you collected. Our prescriptive engine takes active demographic registrations, applies urgency multipliers (1.6x scholars, 1.4x indigent), computes the exact recommended budget per sector, and generates explicit prescriptive directives to close remaining allocation gaps.
4. How does the mobile app stay updated when an admin disburses funds?	We implemented a periodic silent background polling architecture in Flutter (5-second polling timer) that refreshes transaction usage logs and status badges without causing full-screen spinners or resetting scroll positions.
5. How do you prevent canceled or rejected donations from distorting reports?	Canceled or failed transactions are strictly filtered out in both the backend aggregation pipeline and the mobile 'My Contributions' view, ensuring only verified funds in the treasury are counted.